Experimental Protocol and Evaluation Methodology

1 Study Objective

The objective of this experiment is to compare several methods for displaying large volumes of information in Unity, with particular attention given to the impact of networking mechanisms on rendering, computation, and synchronization performance.

2 Experimental Protocol

To ensure the reproducibility of the results, a single protocol is applied to all evaluated solutions.

2.1 Phase 1 – Test Preparation

This phase corresponds to the initialization of the test environment and the setup of all conditions required to launch the experiment.

2.2 Phase 2 – Entity Instantiation

Entities are instantiated progressively in successive waves using identical parameters across all scenarios. This phase evaluates the system's ability to handle the simultaneous appearance of a large number of objects.

2.3 Phase 3 – Entity Movement

Previously instantiated entities are progressively set in motion until 100 % of the simulated population is moving. This phase aims to measure the impact of dynamic simulation on performance.

Note. In the current version of the experiment, the instantiation and movement phases are merged into a single scenario (*Spawn + Move*). The distinction is maintained only to preserve compatibility with previous experiments.

3 Scope of Network Analysis

This study does not focus on the maximum number of simultaneously connected clients. Previous work indicates that performance tends to evolve in a relatively predictable manner as the number of clients increases.

The analysis primarily focuses on:

- data visualization capacity for a single client;
- server-side workload;
- client-side workload;
- network synchronization costs.

4 Experimental Parameters

Parameter	Value
Total number of entities	20 000
Instantiation wave size	400
Delay between waves	10 s
Entities activated per movement step	10 %
Delay before movement activation	5 s
Initial preparation duration	10 s
Waiting time between phases	10 s
Time before automatic shutdown	10 s
Execution mode	Spawn + Move

4.1 Entity Definition

An entity corresponds to a standard Unity cube composed of 12 triangles, using a basic texture and no dynamic lighting.

In networked versions, each entity contains a `NetworkObject` as well as a position synchronization component specific to the implementation being evaluated.

Entities move at a constant speed of 5 m/s along each axis while simultaneously performing a uniform rotation.

5 Study Groups

5.1 Local Configuration

This configuration contains no visible networking layer and is used to evaluate the intrinsic limitations of the rendering engine.

5.2 Networked Configuration

This configuration evaluates server-side and client-side performance behavior within a real client–server architecture.

6 Collected Measurements

6.1 Rendering Performance

- Frames per second (FPS);
- CPU frame time (ms);
- GPU frame time (ms).

6.2 Memory Consumption

- Memory usage (MB).

6.3 Network Performance

- RTT (Round Trip Time);
- RPC-based RTT.

6.4 Network Load

- Upload throughput;
- Download throughput;
- Total volume sent;
- Total volume received;
- Number of packets sent;
- Number of packets received.

6.5 Traffic Analysis

An independent network traffic capture is performed to analyze:

- packets per second;
- exchanged data volume;
- cumulative traffic volumes;
- network load evolution over time.

7 Analysis Method

Metrics are analyzed as a function of the number of entities present in the scene in order to identify:

- server and client limitations;
- the impact of the networking layer;
- performance bottlenecks;
- resource consumption trends;
- hardware limitations of each platform.

8 Client Categories

Two client categories are evaluated:

1. Personal Computer (PC);
2. Meta Quest 3 standalone headset.

Results are analyzed separately for each platform.

9 Hardware Configuration

9.1 Server

The FishNet and Netcode for GameObjects experiments use a dedicated Dell Precision 7550 server running in headless mode.

Component	Specification
Operating System	Windows 11 Enterprise 64-bit
CPU	Intel Core i7-10850H
RAM	32 GB
GPU	NVIDIA Quadro RTX 3000
Graphics API	DirectX 12
Resolution	1920 × 1080

9.2 PC Client

Component	Specification
Operating System	Windows 11 Enterprise 64-bit
CPU	Intel Core Ultra 9 285K
RAM	128 GB
GPU	NVIDIA RTX 5000 Ada Generation
Graphics API	DirectX 12
Resolution	2560 × 1440

9.3 Virtual Reality Client

Immersive tests are performed using a Meta Quest 3 headset.

10 Network Infrastructure

For FishNet and Netcode for GameObjects, the server and clients are connected through a local area network using a Netgear R6020 router.

For Photon, communications are routed through the provider's cloud infrastructure. Consequently, the measured latency values are not directly comparable to those obtained using local architectures.

11 Network Configuration

Local experiments rely on:

- a headless server;
- a PC client or Meta Quest client;
- a Netgear R6020 router;
- stable and controlled network conditions.

In the case of Photon, communications are routed over the Internet and rely on the provider's cloud infrastructure.